

Project Report

Finite Element Solution of the Quantum Harmonic Oscillator

Linear Lagrange Elements, Cubic Hermite Elements, and Variational Convergence

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ABSTRACT

The finite element method (FEM) is a variational way to approximate differential-equation problems with flexible basis functions. This report uses FEM to study the one-dimensional quantum harmonic oscillator. In dimensionless units, the oscillator is described by the potential $V(x) = x^2/2$ and by stationary states satisfying

$$-\frac{1}{2}\psi''(x) + \frac{1}{2}x^2\psi(x) = E\psi(x). \quad (1)$$

The exact eigenvalues are $E_n = n + 1/2$, which makes the problem a useful benchmark rather than an unknown system. It still includes the main ingredients used in larger quantum simulations: a kinetic-energy bilinear form, a potential-energy term, a mass matrix, a normalization constraint, and a generalized matrix eigenvalue problem. Two discretizations are compared: continuous piecewise-linear Lagrange elements and cubic Hermite elements with wavefunction and derivative degrees of freedom. Both methods recover the expected oscillator modes, while the Hermite element reaches much smaller energy errors with fewer elements.

ACKNOWLEDGMENTS

The author acknowledges Ram-Mohan's *Finite Element and Boundary Element Applications in Quantum Mechanics* [1] as the main theoretical inspiration for the variational finite element viewpoint used in this report. The present work follows the same general idea of converting a quantum eigenvalue problem into a finite-dimensional matrix problem, but it presents the calculation using weak-form finite element nota-

tion and focuses on the specific implementation of linear P_1 and C^1 cubic Hermite elements. Numerical computations were performed in Python using NumPy and SciPy, and figures were prepared using Matplotlib. The author is sincerely grateful to Professor Ram-Mohan for his guidance, explanations, and willingness to answer questions throughout this project. His support was instrumental in helping the author understand the finite element approach to quantum-mechanical eigenvalue problems. The author also thanks Arya Eyyappally Retheeshan for helping clarify doubts, discuss difficult concepts, and support the learning process during the development of this report.

INTRODUCTION

The harmonic oscillator is one of the simplest models for motion near a stable equilibrium, and its quantum version is a standard bound-state problem in quantum mechanics. This report uses the one-dimensional quantum harmonic oscillator as a benchmark for the finite element method. The aim is not to solve an unknown problem, since the exact solution is known, but to see how well a finite element approximation reproduces the known energies and wavefunctions.

The one-dimensional quantum harmonic oscillator is described by the quadratic potential

$$V(x) = \frac{1}{2}m\omega^2x^2, \quad (2)$$

where m is the particle mass, ω is the angular frequency, and x is the displacement from equilibrium. The same potential can also be written as $V(x) = \frac{1}{2}kx^2$, with $k = m\omega^2$. The key point

is that the potential grows as $|x|$ increases, so the low-energy quantum states stay localized near the origin and decay in the tails.

The starting point is the stationary Schrödinger equation,

$$\hat{H}\psi = E\psi, \quad (3)$$

where $\hat{H}$ is the Hamiltonian operator, E is the energy eigenvalue, and $\psi(x)$ is the wavefunction. For the harmonic oscillator, the Hamiltonian contains the kinetic energy operator and the quadratic potential energy term:

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2. \quad (4)$$

Substituting this Hamiltonian into the eigenvalue equation gives

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi. \quad (5)$$

This is an eigenvalue problem. The allowed wavefunctions are eigenfunctions of the Hamiltonian, and the corresponding eigenvalues give the allowed energy levels.

For the numerical calculations, it is convenient to use dimensionless oscillator units with $\hbar = m = \omega = 1$. The equation then reduces to

$$-\frac{1}{2}\psi''(x) + \frac{1}{2}x^2\psi(x) = E\psi(x). \quad (6)$$

This nondimensional form is the equation discretized in the finite element calculations.

The harmonic oscillator is especially useful for numerical work because its exact energy spec-

trum is known:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots \quad (7)$$

Because the exact energies and wavefunctions are available, the numerical results can be checked directly. This makes the oscillator a clear test problem for studying finite element accuracy and convergence.

The finite element method represents the continuous wavefunction with basis functions defined on a mesh. In general, the numerical wavefunction is written as

$$\psi_h(x) = \sum_j c_j N_j(x), \quad (8)$$

where $N_j(x)$ are finite element basis functions and c_j are unknown coefficients. This approximation converts the continuous Schrödinger eigenvalue problem into a finite-dimensional matrix eigenvalue problem.

Although the quantum harmonic oscillator is usually introduced through the Hamiltonian eigenvalue equation, Ram-Mohan's finite element treatment emphasizes the equivalent variational formulation. In his harmonic oscillator example, the differential equation is converted into an action integral, and the finite element approximation is obtained by discretizing this action over individual elements. The element matrices are then assembled into a finite-dimensional algebraic eigenvalue problem [1].

The implementation used in this report follows the same general variational motivation, but it is written in a more standard weak-form finite element notation. In this notation, the con-

tinuous Schrödinger equation is converted into matrix form using finite element basis functions, producing a generalized eigenvalue problem

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}. \quad (9)$$

Here, $\mathbf{H}$ is the finite element Hamiltonian matrix, $\mathbf{S}$ is the overlap matrix, and $\mathbf{c}$ is the coefficient vector for the numerical wavefunction. This notation is closely related to Ram-Mohan's action-matrix formulation, but it is not intended to reproduce his harmonic-oscillator derivation line by line.

In particular, the Hermite calculation in this report uses C^1 cubic Hermite elements, with wavefunction values and first derivatives as the degrees of freedom at each node. This should be distinguished from higher-continuity Hermite interpolations, such as formulations that use additional derivative degrees of freedom. The purpose of this report is therefore to compare the implemented linear Lagrange method with the implemented C^1 cubic Hermite method, using the harmonic oscillator as a benchmark problem.

The rest of the report compares linear Lagrange elements with cubic Hermite elements. Linear elements give a simple piecewise-polynomial approximation to the wavefunction, while Hermite elements include derivative degrees of freedom and give a smoother, higher-order approximation. Comparing energies, wavefunctions, and convergence rates against the exact oscillator solution shows how the interpolation choice affects the finite element result.

VARIATIONAL FORMULATION AND NUMERICAL METHOD

After the physical form of the oscillator is introduced, the numerical work is carried out in dimensionless oscillator units. In these units, $\hbar = m = \omega = 1$, and the time-independent Schrödinger equation becomes

$$-\frac{1}{2}\psi''(x) + \frac{1}{2}x^2\psi(x) = E\psi(x). \quad (10)$$

The exact energy levels in these units are

$$E_n = n + \frac{1}{2}, \quad n = 0, 1, 2, \dots \quad (11)$$

The variational structure of the problem can also be expressed through the Rayleigh quotient. For the dimensionless harmonic oscillator, the energy associated with a trial wavefunction ψ is

$$\mathcal{R}[\psi] = \frac{\int_{-L}^L \left[\frac{1}{2} |\psi'(x)|^2 + \frac{1}{2} x^2 |\psi(x)|^2 \right] dx}{\int_{-L}^L |\psi(x)|^2 dx}. \quad (12)$$

The exact eigenfunctions are stationary points of this quotient. In the finite element method, the trial wavefunction is restricted to a finite-dimensional space,

$$\psi_h(x) = \sum_j c_j N_j(x), \quad (13)$$

so the Rayleigh quotient becomes a function of the coefficient vector $\mathbf{c}$:

$$\mathcal{R}[\mathbf{c}] = \frac{\mathbf{c}^T \mathbf{H} \mathbf{c}}{\mathbf{c}^T \mathbf{S} \mathbf{c}}. \quad (14)$$

Requiring this quotient to be stationary with respect to the coefficients produces the generalized

eigenvalue problem

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}. \quad (15)$$

This gives a variational interpretation of the finite element matrices: $\mathbf{H}$ represents the discretized kinetic and potential energy contributions, while $\mathbf{S}$ represents the overlap or normalization matrix.

Ram-Mohan's derivation of the harmonic oscillator finite element equations proceeds through the discretized action integral. In the dimensionless oscillator coordinate, the action contains a derivative term, a quadratic potential term, and an energy-dependent normalization term. When the wavefunction is interpolated over each element, the element action becomes a quadratic form in the nodal unknowns. After the element contributions are assembled, the stationary condition produces a finite-dimensional eigenvalue problem.

The calculation in this report uses the same variational idea but presents it in weak-form notation. Starting from the dimensionless oscillator equation above, the equation is multiplied by a test function $v(x)$ and integrated over the truncated domain. After integrating the second-derivative term by parts, the weak form becomes

$$\begin{aligned} \int_{-L}^L \left[\frac{1}{2} \psi'(x) v'(x) + \frac{1}{2} x^2 \psi(x) v(x) \right] dx \\ = E \int_{-L}^L \psi(x) v(x) dx. \end{aligned} \quad (16)$$

This weak form is not written in exactly the same notation as Ram-Mohan's element action equations, but it leads to the same type of finite-

dimensional matrix problem. The derivative term contributes to the kinetic-energy part of the Hamiltonian matrix, the x^2 term contributes to the potential-energy part, and the right-hand side produces the overlap matrix.

The true harmonic oscillator is defined on the infinite domain $(-\infty, \infty)$. For computation, this domain is replaced by a finite interval,

$$-L \leq x \leq L. \quad (17)$$

Since the harmonic oscillator wavefunctions are localized and decay rapidly for large $|x|$, this truncation is accurate when L is chosen sufficiently large. The boundary conditions used in the finite element calculation are

$$\psi(-L) = 0, \quad \psi(L) = 0. \quad (18)$$

These conditions are not meant to model a physical hard-wall box. They simply approximate the fact that bound-state wavefunctions are already very small near the edges of a sufficiently large domain.

The finite element approximation represents the wavefunction as a linear combination of basis functions:

$$\psi_h(x) = \sum_j c_j N_j(x), \quad (19)$$

where $N_j(x)$ are finite element basis functions and c_j are unknown coefficients. Substituting this approximation into the weak form and choosing the test functions from the same finite element basis gives a generalized matrix eigenvalue problem,

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}. \quad (20)$$

Here, $\mathbf{H}$ is the finite element Hamiltonian matrix, $\mathbf{S}$ is the overlap matrix, and $\mathbf{c}$ is the vector of unknown coefficients.

The weak form determines the entries of these matrices. For basis functions $N_i(x)$ and $N_j(x)$, the Hamiltonian matrix entries are

$$H_{ij} = \int_{-L}^L \left[\frac{1}{2} N_i'(x) N_j'(x) + \frac{1}{2} x^2 N_i(x) N_j(x) \right] dx, \quad (21)$$

and the overlap matrix entries are

$$S_{ij} = \int_{-L}^L N_i(x) N_j(x) dx. \quad (22)$$

Solving the generalized eigenvalue problem produces approximate energy levels and coefficient vectors. Each eigenvalue approximates an oscillator energy, while the corresponding eigenvector determines the finite element approximation to the wavefunction.

The same weak-form framework is used for both element choices. The linear Lagrange method uses piecewise-linear basis functions, while the cubic Hermite method uses higher-order basis functions with both value and derivative degrees of freedom. Comparing the two methods shows how interpolation order and smoothness affect the computed energies and wavefunctions.

LINEAR LAGRANGE ELEMENTS

The first approximation considered here is the linear Lagrange element, also called the P_1 element. In this method, the computational interval $[-L, L]$ is divided into N elements with

nodes

$$x_0, x_1, \dots, x_N. \quad (23)$$

Each element e is the interval

$$[x_e, x_{e+1}], \quad (24)$$

with element length

$$h_e = x_{e+1} - x_e. \quad (25)$$

Within each element, the wavefunction is approximated by a linear function. Thus, over the element $[x_e, x_{e+1}]$, the finite element approximation has the form

$$\psi_h(x) = c_e N_1^{(e)}(x) + c_{e+1} N_2^{(e)}(x), \quad (26)$$

where c_e and c_{e+1} are the unknown nodal values of the wavefunction, and $N_1^{(e)}(x)$ and $N_2^{(e)}(x)$ are the local linear basis functions.

The local basis functions are chosen so that each function is equal to one at one endpoint of the element and zero at the other endpoint. They are given by

$$N_1^{(e)}(x) = \frac{x_{e+1} - x}{h_e}, \quad N_2^{(e)}(x) = \frac{x - x_e}{h_e}. \quad (27)$$

These functions interpolate the wavefunction between neighboring nodes. Because adjacent elements share nodal values, the global approximation is continuous across the full domain. Since each element still uses its own linear polynomial, the derivative is generally discontinuous at element boundaries.

Using the local linear basis functions, the element matrices can be written explicitly. For an element $e = [x_e, x_{e+1}]$ of length $h_e = x_{e+1} - x_e$,

the kinetic-energy contribution is

$$\mathbf{K}^{(e)} = \frac{1}{2h_e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \quad (28)$$

This matrix comes from the derivative term

$$\frac{1}{2} \int_{x_e}^{x_{e+1}} \frac{dN_i^{(e)}}{dx} \frac{dN_j^{(e)}}{dx} dx. \quad (29)$$

The overlap matrix for the same element is

$$\mathbf{S}^{(e)} = \frac{h_e}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. \quad (30)$$

This matrix comes from the integral of the product of basis functions,

$$S_{ij}^{(e)} = \int_{x_e}^{x_{e+1}} N_i^{(e)}(x) N_j^{(e)}(x) dx. \quad (31)$$

For the harmonic oscillator, the potential matrix is

$$V_{ij}^{(e)} = \int_{x_e}^{x_{e+1}} \frac{1}{2} x^2 N_i^{(e)}(x) N_j^{(e)}(x) dx. \quad (32)$$

In the implementation this integral is evaluated by Gauss–Legendre quadrature. If the local coordinate $t = (x - x_e)/h_e$ is used, so that $x = x_e + h_e t$, this potential contribution may also be written explicitly as

$$\mathbf{V}^{(e)} = \frac{h_e}{2} \begin{bmatrix} \frac{x_e^2}{3} + \frac{x_e h_e}{6} + \frac{h_e^2}{30} & \frac{x_e^2}{6} + \frac{x_e h_e}{6} + \frac{h_e^2}{20} \\ \frac{x_e^2}{6} + \frac{x_e h_e}{6} + \frac{h_e^2}{20} & \frac{x_e^2}{3} + \frac{x_e h_e}{2} + \frac{h_e^2}{5} \end{bmatrix}$$

Therefore, the local Hamiltonian matrix is

$$\mathbf{H}^{(e)} = \mathbf{K}^{(e)} + \mathbf{V}^{(e)}. \quad (33)$$

The global matrices $\mathbf{H}$ and $\mathbf{S}$ are obtained by

assembling these local element matrices over the full mesh. The overlap matrix appears because the finite element basis functions are not generally orthonormal, so the final discrete problem is a generalized eigenvalue problem rather than an ordinary eigenvalue problem.

After the local element matrices are computed, they are assembled into global matrices by adding each element contribution to the entries associated with its two global nodes. This gives the finite-dimensional system

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}, \quad (34)$$

where $\mathbf{H}$ is the assembled Hamiltonian matrix, $\mathbf{S}$ is the assembled overlap matrix, and $\mathbf{c}$ is the vector of unknown nodal coefficients.

The boundary conditions are imposed by setting the wavefunction to zero at the endpoints of the truncated domain:

$$\psi(-L) = 0, \quad \psi(L) = 0. \quad (35)$$

For the linear element approximation, this means that the coefficients associated with the first and last nodes are constrained to be zero. The remaining interior coefficients are then determined by solving the generalized eigenvalue problem.

The linear Lagrange method is the simplest finite element approximation used in this report. It gives a useful baseline because it captures the basic finite element structure while using only nodal wavefunction values as degrees of freedom. Since the approximation is only piecewise linear, a fairly fine mesh is needed to represent curved or oscillatory wavefunctions, especially for higher-energy states.

CUBIC HERMITE ELEMENTS

The second approximation used in this report is a C^1 cubic Hermite element. This is the Hermite element implemented in the numerical code: each node has two degrees of freedom, the value of the wavefunction and its first derivative. Thus, at each node x_i , the unknowns are

$$\psi(x_i) \quad \text{and} \quad \psi'(x_i). \quad (36)$$

These degrees of freedom determine a cubic polynomial on each element and produce an approximation whose value and first derivative are continuous across element boundaries. This should be distinguished from higher-continuity Hermite interpolations that use additional derivative degrees of freedom.

Consider an element e on the interval $[x_e, x_{e+1}]$, with element length

$$h_e = x_{e+1} - x_e. \quad (37)$$

A local coordinate t is introduced by

$$t = \frac{x - x_e}{h_e}, \quad 0 \leq t \leq 1. \quad (38)$$

Using this local coordinate, the cubic Hermite approximation on the element is written as

$$\begin{aligned} \psi_h(x) = & u_e \mathcal{H}_1(t) + d_e \mathcal{H}_2(t) \\ & + u_{e+1} \mathcal{H}_3(t) + d_{e+1} \mathcal{H}_4(t), \end{aligned} \quad (39)$$

where $u_e = \psi(x_e)$, $d_e = \psi'(x_e)$, $u_{e+1} = \psi(x_{e+1})$, and $d_{e+1} = \psi'(x_{e+1})$. The four cubic Hermite

basis functions are

$$\mathcal{H}_1(t) = 2t^3 - 3t^2 + 1, \quad (40)$$

$$\mathcal{H}_2(t) = h_e(t^3 - 2t^2 + t), \quad (41)$$

$$\mathcal{H}_3(t) = -2t^3 + 3t^2, \quad (42)$$

$$\mathcal{H}_4(t) = h_e(t^3 - t^2). \quad (43)$$

The functions $\mathcal{H}_1(t)$ and $\mathcal{H}_3(t)$ set the wavefunction value at the left and right endpoints, while $\mathcal{H}_2(t)$ and $\mathcal{H}_4(t)$ set the derivative at those endpoints.

The reason for using a cubic polynomial is that four local degrees of freedom are available on each element: two nodal values and two nodal derivatives. A cubic is the lowest-degree polynomial that can satisfy these four interpolation conditions. This gives a higher-order approximation without claiming to reproduce every detail of Ram-Mohan's Hermite formulation.

Because neighboring elements share both nodal values and derivative degrees of freedom, the Hermite approximation has C^1 continuity. The approximate wavefunction and its first derivative are continuous across the computational domain. This added smoothness is the main difference from the linear Lagrange approximation, whose derivative is generally discontinuous at element boundaries.

As in the linear case, the Hermite element matrices are obtained by substituting the finite element approximation into the weak form of the dimensionless oscillator problem. Each Hermite element has four local basis functions, so the local Hamiltonian and overlap matrices are 4×4 .

The local Hamiltonian matrix is

$$H_{ij}^{(e)} = \int_{x_e}^{x_{e+1}} \left[\frac{1}{2} \frac{d\mathcal{H}_i}{dx} \frac{d\mathcal{H}_j}{dx} + \frac{1}{2} x^2 \mathcal{H}_i(x) \mathcal{H}_j(x) \right] dx, \quad (44)$$

and the local overlap matrix is

$$S_{ij}^{(e)} = \int_{x_e}^{x_{e+1}} \mathcal{H}_i(x) \mathcal{H}_j(x) dx, \quad (45)$$

where $i, j \in \{1, 2, 3, 4\}$. The first term in $H_{ij}^{(e)}$ represents the kinetic energy contribution, while the second term represents the contribution from the harmonic oscillator potential.

The local Hermite matrices are assembled into the global Hamiltonian matrix $\mathbf{H}$ and overlap matrix $\mathbf{S}$ in the same way as for the linear elements. Because each node has two degrees of freedom, the global coefficient vector contains both nodal wavefunction values and nodal derivative values. The resulting discrete problem again has the generalized eigenvalue form

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}. \quad (46)$$

The boundary conditions are imposed on the value of the wavefunction at the endpoints of the truncated domain:

$$\psi(-L) = 0, \quad \psi(L) = 0. \quad (47)$$

The derivative degrees of freedom at the endpoints are not fixed by these Dirichlet boundary conditions. They remain part of the unknown coefficient vector and are determined by the eigenvalue problem.

The cubic Hermite method used here is

smoother and higher order than the linear Lagrange method. Since harmonic oscillator wavefunctions are smooth, derivative degrees of freedom help the numerical approximation capture curvature more efficiently than piecewise-linear elements. The Hermite calculation in this report should therefore be understood specifically as a C^1 cubic Hermite implementation, not as the same Hermite oscillator calculation presented in Ram-Mohan's table. The comparison here is between the implemented P_1 linear method and the implemented C^1 cubic Hermite method.

RESULTS AND CONVERGENCE

The results in this section come from the finite element implementation described above, not from a direct reproduction of Ram-Mohan's harmonic-oscillator tables. The numerical implementation used the dimensionless Hamiltonian

$$\hat{H} = -\frac{1}{2} \frac{d^2}{dx^2} + \frac{1}{2} x^2, \quad (48)$$

with exact eigenvalues $E_n = n + \frac{1}{2}$. The infinite physical domain was truncated to the finite interval $[-8, 8]$, and homogeneous Dirichlet boundary conditions were applied to the wavefunction values at the endpoints.

For the linear Lagrange calculation, the domain was divided into $N = 300$ elements. For the cubic Hermite calculation, the domain was divided into $N = 60$ elements. Although the Hermite calculation used fewer elements, each node carried both a wavefunction value and a derivative degree of freedom. Element integrals were evaluated using Gauss–Legendre quadrature, and the assembled generalized eigenvalue

problem

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c} \quad (49)$$

was solved using a symmetric eigensolver to obtain the numerical energy levels and wavefunctions.

The computed results were then compared with the exact harmonic oscillator energies and wavefunctions. This comparison allows the error in both the eigenvalues and the wavefunction shapes to be measured directly.

Energy Eigenvalues

Table 1 compares the first ten computed energy levels with the exact harmonic oscillator spectrum. Both finite element methods recover the expected evenly spaced spectrum, but the Hermite method gives much smaller relative errors while using fewer elements.

The linear method is accurate, although its error grows gradually for higher energy states. This is expected because higher oscillator states have more oscillations and are harder to represent with piecewise-linear functions. The Hermite method stays more accurate over the same range. For example, for the ground state $n = 0$, the linear relative error is 1.778×10^{-4} , while the Hermite relative error is 7.068×10^{-8} . For $n = 9$, the linear relative error increases to 1.691×10^{-3} , while the Hermite relative error remains at 1.594×10^{-5} .

Figure 1 gives a visual view of the energy spectrum. Since the errors are small compared with the energy scale, the exact, linear FEM, and Hermite FEM bars nearly overlap. The numerical table is therefore the clearest way to see the difference between the two methods.

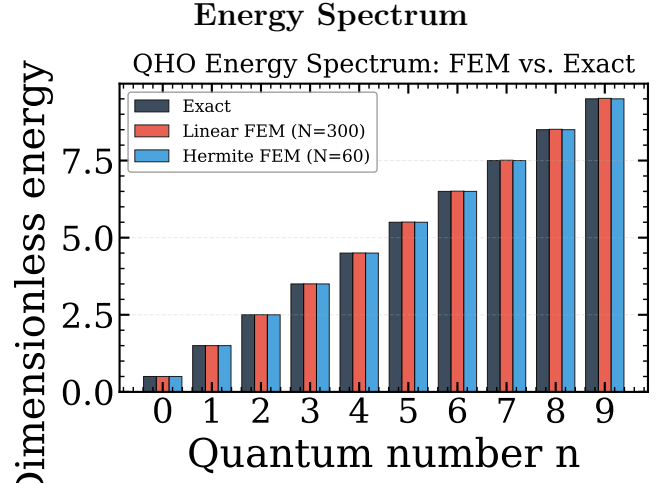


Figure 1: Comparison of the exact harmonic oscillator energy spectrum with the linear and Hermite finite element approximations for $n = 0$ through $n = 9$.

Wavefunction Accuracy

Figures 2 and 3 compare the computed wavefunctions with the exact harmonic oscillator wavefunctions for the first six states. The linear finite element approximation is expected to work well for the low oscillator states, but the approximation becomes more demanding as the quantum number increases. Higher states have more nodes and more rapid oscillations, so a piecewise-linear basis needs a fine mesh to capture the changing curvature accurately.

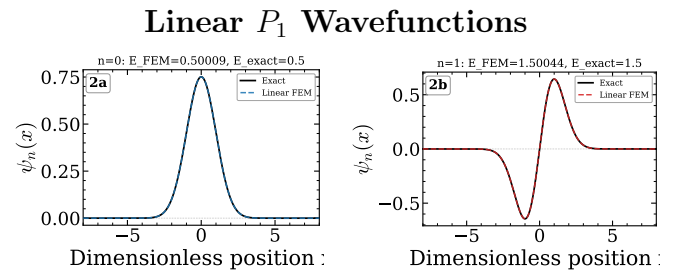


Table 1: Comparison of exact and finite element energy eigenvalues for the first ten harmonic oscillator states. The relative error is defined as $|E_{\text{FEM}} - E_{\text{exact}}|/E_{\text{exact}}$.

n	E_{exact}	E_{linear}	Relative Error	E_{Hermite}	Relative Error
0	0.50000	0.50008888	1.778×10^{-4}	0.50000004	7.068×10^{-8}
1	1.50000	1.50044434	2.962×10^{-4}	1.50000031	2.070×10^{-7}
2	2.50000	2.50115506	4.620×10^{-4}	2.50000138	5.503×10^{-7}
3	3.50000	3.50222086	6.345×10^{-4}	3.50000420	1.200×10^{-6}
4	4.50000	4.50364156	8.092×10^{-4}	4.50001012	2.249×10^{-6}
5	5.50000	5.50541695	9.849×10^{-4}	5.50002078	3.779×10^{-6}
6	6.50000	6.50754686	1.161×10^{-3}	6.50003807	5.858×10^{-6}
7	7.50000	7.51003109	1.337×10^{-3}	7.50006409	8.545×10^{-6}
8	8.50000	8.51286946	1.514×10^{-3}	8.50010108	1.189×10^{-5}
9	9.50000	9.51606176	1.691×10^{-3}	9.50015144	1.594×10^{-5}

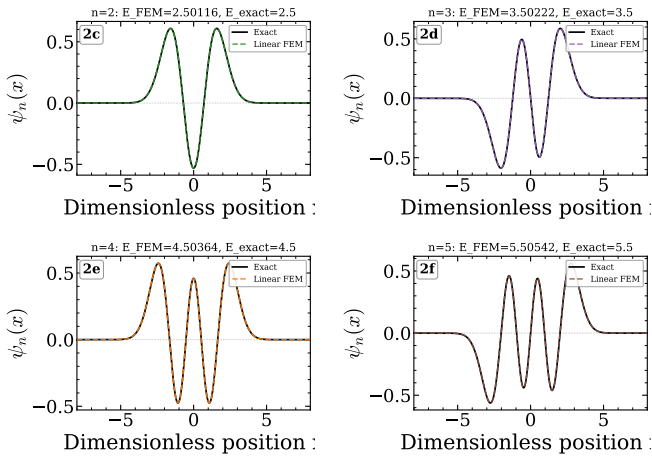


Figure 2: Comparison of exact wavefunctions with the linear P_1 finite element approximation for $n = 0$ through $n = 5$, using $N = 300$ elements and $L = 8$. Panels 2a through 2f correspond to $n = 0$ through $n = 5$.

The linear FEM approximation captures the overall shape and nodal structure of each exact wavefunction. The curves overlap closely at the scale of the plots, so the increasing error for higher states is clearer in the numerical energy table than in the wavefunction overlays alone.

The Hermite approximation gives a closer match with fewer elements. This happens because the cubic Hermite basis includes both function values and derivative degrees of freedom, so each element can represent curvature more accu-

rately. In the plotted states, the Hermite FEM curves are nearly indistinguishable from the exact wavefunctions at the scale of the figure.

Cubic Hermite Wavefunctions

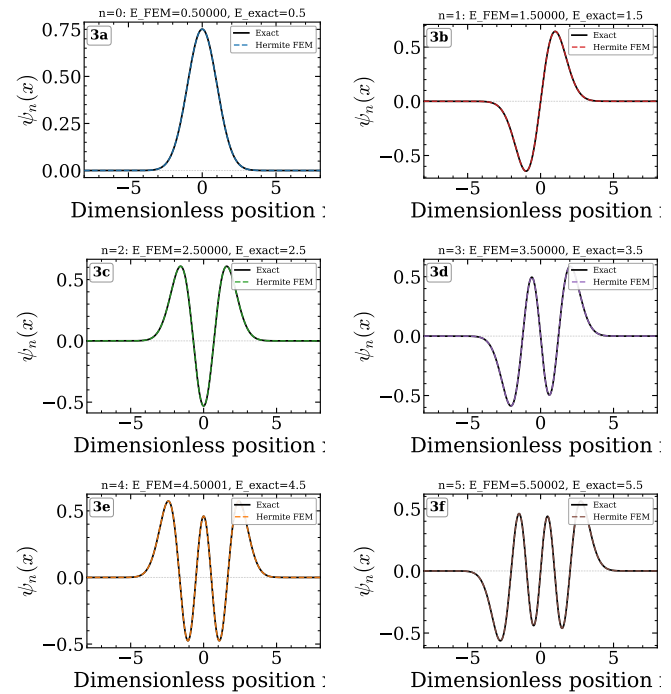


Figure 3: Comparison of exact wavefunctions with the cubic Hermite finite element approximation for $n = 0$ through $n = 5$, using $N = 60$ elements and $L = 8$. Panels 3a through 3f correspond to $n = 0$ through $n = 5$.

The Hermite curves are visually almost indistinguishable from the exact wavefunctions

at the scale of the plots. This agreement is achieved with fewer elements than the linear calculation because the Hermite basis functions include derivative degrees of freedom and can represent curvature more accurately within each element.

Figure 4 shows the pointwise absolute error between the numerical and exact wavefunctions for selected states. These absolute-error plots make the difference between the methods easier to see than the wavefunction overlays alone.

Wavefunction Error Comparison

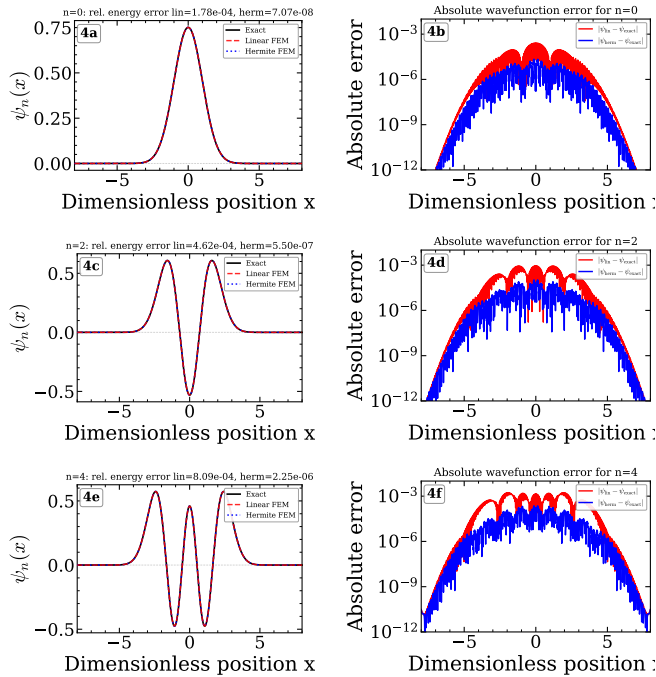


Figure 4: Wavefunction and pointwise absolute error comparison for selected states. The left panels compare exact and finite element wavefunctions, while the right panels show $|\psi_{\text{FEM}}(x) - \psi_{\text{exact}}(x)|$ for the linear and Hermite approximations.

The absolute-error panels reveal differences that are difficult to see in the overlays. The Hermite approximation generally gives smaller pointwise errors than the linear approximation,

especially in the central region where the wavefunction has significant amplitude. Errors in the far tails should be interpreted carefully because the exact wavefunction is extremely small there.

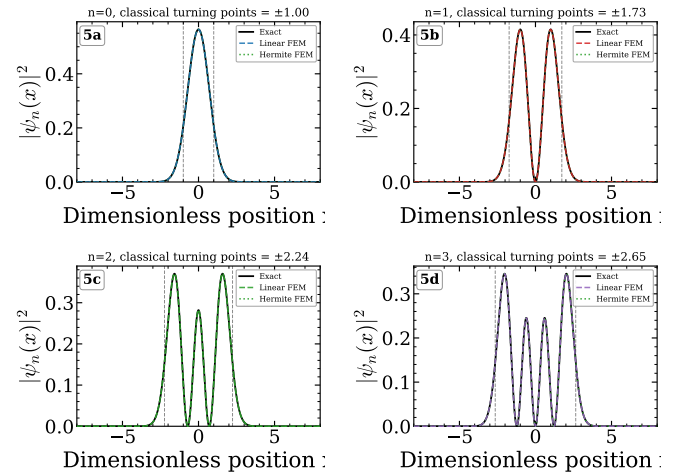
Probability Densities

The probability density $|\psi_n(x)|^2$ describes the spatial distribution of the n -th oscillator state. Figure 5 compares the exact, linear FEM, and Hermite FEM probability densities for $n = 0$ through $n = 5$. The dashed vertical lines mark the classical turning points,

$$x_{\text{cl}} = \pm \sqrt{2E_n}. \quad (50)$$

As n increases, the probability density spreads over a wider region and develops more internal oscillations. Both finite element methods reproduce these features, while the Hermite approximation follows the exact density more closely because of its higher-order smoothness.

Probability Densities



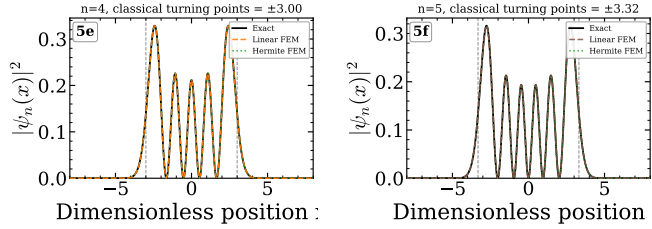


Figure 5: Probability densities $|\psi_n(x)|^2$ for the first six harmonic oscillator states. Panels 5a through 5f correspond to $n = 0$ through $n = 5$. The dashed vertical lines indicate the classical turning points $x_{cl} = \pm\sqrt{2E_n}$.

The probability-density plots add a physical interpretation to the wavefunction comparison. As n increases, the density spreads farther from the origin and develops more internal peaks. The classical turning points move outward because higher-energy states explore a larger region of the harmonic potential.

Convergence Under Mesh Refinement

A useful test of a numerical method is whether its error decreases as the mesh is refined. Figure 6 shows the relative energy error as a function of element size h for the first four states. The linear P_1 element results follow approximately second-order convergence, while the cubic Hermite element results follow approximately fourth-order convergence.

This convergence behavior explains the difference in accuracy seen in Table 1. The slopes support approximately $\mathcal{O}(h^2)$ behavior for the linear method and approximately $\mathcal{O}(h^4)$ behavior for the Hermite method. This is one of the strongest pieces of evidence that the Hermite approximation is not just slightly more accurate, but systematically higher order for this smooth eigenvalue problem.

Mesh-Refinement Convergence

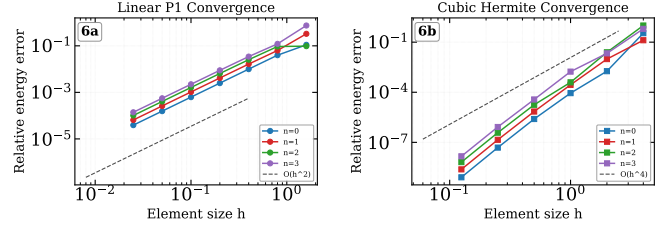


Figure 6: Eigenvalue convergence under h -refinement for linear P_1 and cubic Hermite elements. The linear method follows approximately $\mathcal{O}(h^2)$ behavior, while the Hermite method follows approximately $\mathcal{O}(h^4)$ behavior.

Sensitivity to Domain Truncation

The exact harmonic oscillator is defined on the infinite domain $(-\infty, \infty)$, while the finite element calculation uses a finite interval $[-L, L]$. Figure 7 shows how the relative energy error changes as the cutoff L is varied while the number of elements is held fixed.

Domain-Truncation Sensitivity

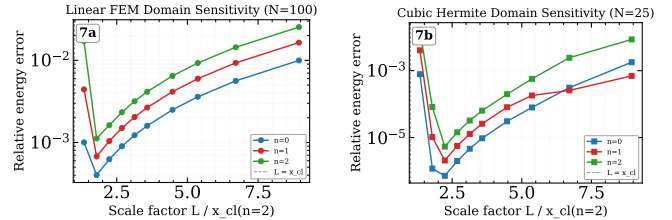


Figure 7: Sensitivity of the relative energy error to the domain cutoff L at fixed element count. The horizontal axis is scaled by the classical turning point for the $n = 2$ state.

The choice of L involves a balance. If L is too small, the finite boundary cuts into the wavefunction tails and distorts the computed energy. If L is too large while the number of elements is fixed, the element size increases and the mesh becomes too coarse near the region where the wavefunction is largest. Accurate convergence

therefore requires a domain large enough to contain the decaying tails, while still keeping enough mesh resolution near the origin.

Finite Element Shape Functions

Figure 8 illustrates the basis functions used by the two finite element methods. The linear P_1 hat functions control only the wavefunction values at nodes. These basis functions produce a continuous wavefunction, but their derivatives are generally discontinuous between elements. The cubic Hermite basis functions use both value and derivative degrees of freedom, allowing the approximation to represent both the wavefunction and its slope continuously across element boundaries.

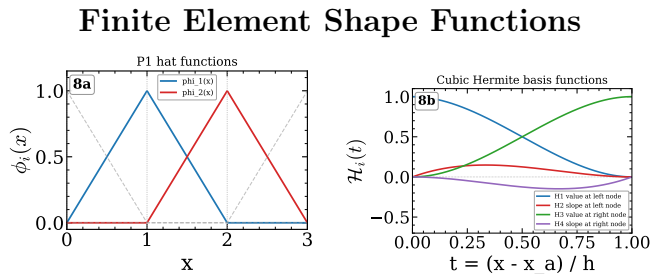


Figure 8: Comparison of the finite element basis functions used in this report. Panel 8a shows the linear P_1 hat functions, while Panel 8b shows the cubic Hermite basis functions.

Overall, both finite element methods reproduce the oscillator spectrum and wavefunctions. The cubic Hermite element is more accurate than the linear Lagrange element for this smooth bound-state problem. The energy table, wavefunction comparisons, absolute-error plots, and convergence study support the same conclusion: higher-order smooth interpolation improves the finite element approximation.

DISCUSSION

Interpretation of the Energy Errors

The computed spectrum confirms that both discretizations approximate the harmonic oscillator eigenvalues, although the corresponding error levels are quite different. For the linear P_1 approximation, the eigenvalues remain close to the exact values over the states considered. The relative error increases with the quantum number n , which is consistent with the structure of the oscillator eigenfunctions. Higher states contain more nodes, stronger curvature changes, and shorter local length scales. A piecewise-linear trial space can represent these features accurately only when the mesh is fine enough to resolve them.

The cubic Hermite method gives substantially smaller energy errors, even though it uses fewer elements in the calculation. This improvement follows from the richer local approximation space. Because each node carries both a wavefunction value and a derivative degree of freedom, the element polynomial can reproduce curvature more effectively than a straight-line interpolant. The energy table shows that there is an actual gain in approximation efficiency from using the cubic Hermite basis.

The energy spectrum plot should be interpreted mainly as a qualitative check. The nearly overlapping bars show that both methods recover the expected evenly spaced oscillator spectrum. Since the energy errors are small relative to the scale of the eigenvalues, the numerical table and convergence plots provide the more useful accuracy comparison.

Effect of Element Smoothness

From the finite element point of view, the difference between the methods is closely tied to the regularity of the trial space. Linear Lagrange elements enforce continuity of the approximate wavefunction across element boundaries, while the derivative remains discontinuous in general. The resulting trial functions are continuous and piecewise linear, with slopes that change from one element to the next. This space is admissible in the weak form of the Schrödinger equation, since the formulation only requires first derivatives in an integral sense. For a smooth bound-state wavefunction, however, the limited regularity reduces approximation efficiency.

Cubic Hermite elements use a different set of degrees of freedom. By including both nodal values and nodal derivatives, they produce a C^1 approximation in which the wavefunction and its first derivative are continuous across element boundaries. This smoothness is well matched to the harmonic oscillator eigenfunctions, which are analytic functions on the real line. The smaller errors and faster convergence of the Hermite calculation are therefore consistent with the regularity of the exact solution.

Linear elements remain useful because they are simple, robust, and often effective. In the present smooth one-dimensional eigenvalue problem, the additional smoothness and higher polynomial order of the Hermite basis provide a clear numerical advantage.

Role of Domain Truncation

The harmonic oscillator is posed on the infinite interval $(-\infty, \infty)$. The finite element calcu-

lation uses a bounded computational domain instead. Replacing the infinite interval by $[-L, L]$, together with homogeneous Dirichlet boundary conditions at $x = -L$ and $x = L$, approximates the decay of the bound-state wavefunctions. This approximation is accurate when the chosen boundary lies far enough into the classically forbidden region that the wavefunction values are already small.

The domain sensitivity results show that the choice of L involves two competing effects. If L is too small, the artificial boundary removes part of the wavefunction tail and perturbs the energy. If L is too large while the number of elements is fixed, the element size increases and the mesh loses resolution near the region where the wavefunction has its largest amplitude and most important structure. The best choice of domain therefore balances tail containment against mesh resolution.

The same issue appears in many quantum finite element calculations. Infinite or very large physical domains are often replaced by finite computational domains. The accuracy of the result then depends on both the placement of the artificial boundary and the distribution of mesh points inside that boundary.

Relationship to Ram-Mohan's Formulation

Ram-Mohan's treatment of the harmonic oscillator develops the finite element equations through an action-integral formulation. In that approach, the approximate wavefunction is substituted into the element action, and stationarity of the assembled action gives a finite-dimensional

eigenvalue problem. The present report uses the same variational basis, but expresses the calculation in weak-form matrix notation,

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}.$$

These formulations are closely related. Both impose the stationary condition of the quantum eigenvalue problem after restricting the wavefunction to a finite-dimensional trial space.

The Hermite calculation in this report should also be interpreted with some care. Ram-Mohan discusses Hermite interpolation in a broader finite element setting, including higher-continuity possibilities. The implementation studied here is specifically a C^1 cubic Hermite discretization with wavefunction and first-derivative degrees of freedom at each node. The numerical comparison is between the two discretizations implemented in this report: linear Lagrange and cubic Hermite elements.

EXTENSIONS AND FUTURE WORK

Exploratory Quintic Hermite Extension with C^2 Continuity

The cubic Hermite calculation used in the main part of this report is a C^1 approximation: the wavefunction and its first derivative are continuous across element boundaries. A natural higher-order extension is a quintic Hermite element with C^2 continuity. In this element, each node carries three degrees of freedom,

$$\psi(x_i), \quad \psi'(x_i), \quad \psi''(x_i).$$

An element with two endpoints therefore has six local interpolation conditions. Since a quintic polynomial has six coefficients, it is the lowest-degree polynomial that can match the value, first derivative, and second derivative at both endpoints.

For an element $[x_e, x_{e+1}]$ of length h_e , define the local coordinate

$$t = \frac{x - x_e}{h_e}, \quad 0 \leq t \leq 1.$$

The quintic Hermite approximation can then be written as

$$\begin{aligned} \psi_h(x) = & u_e Q_1(t) + d_e Q_2(t) + a_e Q_3(t) \\ & + u_{e+1} Q_4(t) + d_{e+1} Q_5(t) + a_{e+1} Q_6(t), \end{aligned} \quad (51)$$

where

$$u_e = \psi(x_e), \quad d_e = \psi'(x_e), \quad a_e = \psi''(x_e),$$

with corresponding definitions at the right endpoint. The six local basis functions are

$$Q_1(t) = 1 - 10t^3 + 15t^4 - 6t^5, \quad (52)$$

$$Q_2(t) = h_e (t - 6t^3 + 8t^4 - 3t^5), \quad (53)$$

$$Q_3(t) = \frac{h_e^2}{2} (t^2 - 3t^3 + 3t^4 - t^5), \quad (54)$$

$$Q_4(t) = 10t^3 - 15t^4 + 6t^5, \quad (55)$$

$$Q_5(t) = h_e (-4t^3 + 7t^4 - 3t^5), \quad (56)$$

$$Q_6(t) = \frac{h_e^2}{2} (t^3 - 2t^4 + t^5). \quad (57)$$

The factors of h_e and h_e^2 ensure that the derivative and curvature degrees of freedom are defined with respect to the physical coordinate x , rather than the dimensionless coordinate t .

The basis functions are constructed so that Q_1 , Q_2 , and Q_3 control the value, slope, and curvature at the left endpoint, while Q_4 , Q_5 , and Q_6 control the corresponding quantities at the right endpoint.

Quintic C^2 Hermite Basis Functions

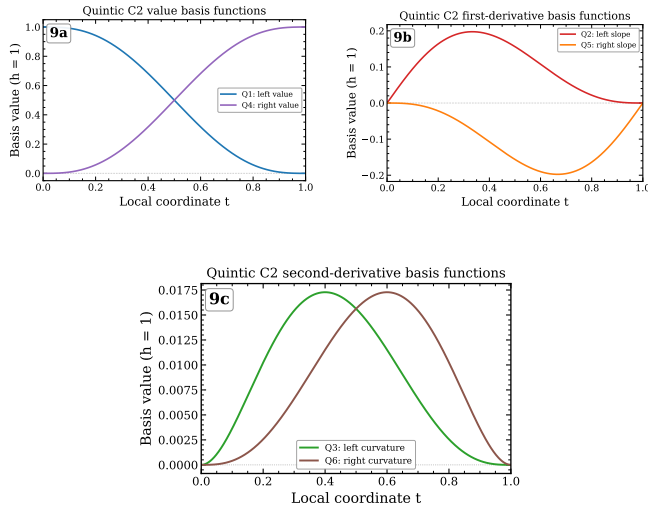


Figure 9: Quintic C^2 Hermite basis functions. Panels 9a, 9b, and 9c show the basis functions associated with endpoint value, first-derivative, and second-derivative degrees of freedom, respectively.

Because adjacent elements share the value, first derivative, and second derivative at each node, the quintic approximation is C^2 across element boundaries. This continuity is higher than the present weak form requires, since the weak form only involves first derivatives. The exploratory comparison therefore reflects the combined effect of quintic polynomial order, C^2 continuity, and the larger local interpolation space, rather than isolating continuity alone.

The weak-form assembly used for the linear and cubic bases extends directly to the quintic basis. Each element produces 6×6 Hamiltonian and overlap matrices, while the Dirichlet condi-

tions still constrain only the endpoint values,

$$\psi(-L) = 0, \quad \psi(L) = 0.$$

The derivative and curvature degrees of freedom at the two boundaries remain free.

Comparison at Comparable Degrees of Freedom

An exploratory calculation used the same truncated domain, $[-8, 8]$, with approximately equal numbers of free degrees of freedom. The linear calculation used 121 elements and 120 free degrees of freedom, the cubic C^1 Hermite calculation used 60 elements and 120 free degrees of freedom, and the quintic C^2 Hermite calculation used 40 elements and 121 free degrees of freedom.

Table 2 shows that the quintic approximation gives substantially smaller energy errors for the selected states at a comparable number of free degrees of freedom. The difference is especially clear for the $n = 4$ state, which has several nodes and therefore provides a more demanding interpolation test than the ground state.

Comparison at Comparable Degrees of Freedom

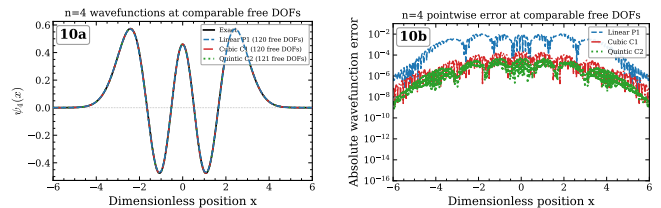


Figure 10: Comparison of the $n = 4$ harmonic oscillator state at approximately equal numbers of free degrees of freedom. Panel 10a shows the numerical wavefunctions, which overlap closely at the plotted scale. Panel 10b shows the pointwise absolute error and reveals the improved accuracy of the cubic and quintic Hermite approximations.

Table 2: Exploratory comparison of the three finite element spaces at approximately equal numbers of free degrees of freedom.

Method	Elements	Free DOFs	E_0	Relative Error in E_0	E_4	Relative Error in E_4
Linear P_1	121	120	0.500546239059	1.092×10^{-3}	4.522293581103	4.954×10^{-3}
Cubic C^1 Hermite	60	120	0.500000035339	7.068×10^{-8}	4.500010121251	2.249×10^{-6}
Quintic C^2 Hermite	40	121	0.500000000002	4.821×10^{-12}	4.500000003705	8.234×10^{-10}

The wavefunction curves in Figure 10 overlap closely because all three methods represent the overall shape of the state. The absolute-error plot is therefore more informative than the wavefunction overlay alone. It shows that the cubic C^1 method improves substantially on the linear approximation and that the quintic C^2 calculation reduces the error further for this smooth oscillator state.

Convergence of the Three Element Types

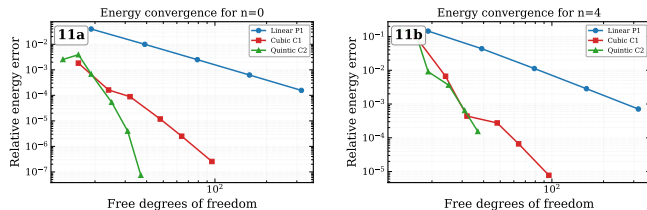


Figure 11: Relative energy error as a function of the number of free degrees of freedom for the three discretizations. Panel 11a shows the ground state, while Panel 11b shows the more oscillatory $n = 4$ state.

Figure 11 supports the same conclusion over multiple mesh sizes. The linear P_1 approximation converges most slowly, the cubic C^1 Hermite approximation improves more rapidly, and the quintic C^2 calculation reaches smaller errors with comparatively few degrees of freedom. This extension should be interpreted as an exploratory comparison of the implemented element spaces, not as a direct reproduction of a particular higher-order Hermite table in Ram-

Mohan's text.

Additional Future Directions

The one-dimensional harmonic oscillator is a useful benchmark because its exact solution is known, but the same finite element framework can be extended to more general quantum systems. These extensions would use the same basic structure developed in this report: choose a finite-dimensional trial space, assemble the Hamiltonian and overlap matrices, impose appropriate boundary conditions, and solve the resulting algebraic problem.

One natural extension is the two-dimensional quantum harmonic oscillator. The isotropic two-dimensional oscillator is relevant as a simple model of parabolic confinement, including idealized models of quantum dots. In two dimensions, the wavefunction depends on two spatial variables rather than one. The finite element basis functions may be formed using tensor-product basis functions or by using two-dimensional elements directly. The element integrals become area integrals, but the variational structure remains the same. The kinetic-energy term, potential-energy term, and overlap term are still assembled element by element into a generalized eigenvalue problem. Thus, the two-dimensional oscillator would be a natural next test of the same method.

Another extension is the anharmonic oscillator. Instead of using only the quadratic potential

$$V(x) = \frac{1}{2}x^2,$$

one could add a quartic term,

$$V(x) = \frac{1}{2}x^2 + \lambda x^4.$$

For nonzero λ , the energy spectrum is no longer given by the simple harmonic oscillator formula $E_n = n + 1/2$. This makes the problem a more realistic numerical test, because the finite element method would be used to approximate a system without a simple closed-form spectrum. In the finite element formulation, this modification mainly changes the potential-energy matrix. The kinetic-energy and overlap matrices keep the same structure, while the potential contribution is recomputed using the new function $V(x)$. Gauss–Legendre quadrature can evaluate these element integrals accurately, and for polynomial integrands it can be exact when the quadrature order is chosen sufficiently high.

A third extension is to semiconductor heterostructures or quantum wells. In these systems, the effective mass may vary with position, so the constant mass m is replaced by a position-dependent effective mass $m^*(x)$. The kinetic-energy matrix then contains the material parameter inside the element integral. In one dimension, a typical kinetic-energy contribution has the form

$$K_{ij} = \frac{\hbar^2}{2} \int N'_i(x) [m^*(x)]^{-1} N'_j(x) \, dx.$$

This expression can still be evaluated element

by element. The main difference is that the coefficient multiplying the derivative terms is no longer constant. This illustrates one advantage of finite element methods in quantum mechanics: spatially varying material parameters and nonuniform device structures can be included directly in the assembled matrices.

The method could also be extended to time-dependent quantum problems. Instead of solving only the stationary eigenvalue problem, the finite element basis could be used to approximate the time-dependent Schrödinger equation. This would produce a semidiscrete system of ordinary differential equations for the time-dependent coefficient vector. Time integration methods such as Crank–Nicolson or exponential time integrators could then be used to evolve the wavefunction. Such an extension would make it possible to study wavepacket motion, tunneling behavior, and response to time-dependent potentials.

Finally, future work could examine higher-order finite element spaces and p -refinement. In this report, the comparison was between linear P_1 Lagrange elements and C^1 cubic Hermite elements. Another approach would be to keep the mesh fixed while increasing the polynomial degree inside each element. For smooth eigenfunctions, higher-order or spectral-type elements can converge very rapidly. Since the harmonic oscillator eigenfunctions are smooth, this problem would be a natural setting for comparing h -refinement, which reduces the element size, with p -refinement, which increases the polynomial order.

These extensions show that the harmonic oscillator calculation is a starting point rather

than an endpoint. The same finite element ideas used here can be applied to higher-dimensional systems, non-quadratic potentials, position-dependent material parameters, and time-dependent quantum dynamics.

CONCLUSION

This report applied the finite element method to the one-dimensional quantum harmonic oscillator. Because the exact eigenvalues and eigenfunctions are known, the oscillator provides a useful benchmark for assessing both energy accuracy and wavefunction accuracy. The comparison can therefore be made quantitatively.

In dimensionless oscillator units, the problem was written as

$$-\frac{1}{2}\psi''(x) + \frac{1}{2}x^2\psi(x) = E\psi(x).$$

The weak and variational formulation led to the generalized matrix eigenvalue problem

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c},$$

where $\mathbf{H}$ is the finite element Hamiltonian matrix, $\mathbf{S}$ is the overlap matrix, and $\mathbf{c}$ contains the expansion coefficients of the approximate wavefunction.

Two finite element spaces were compared. The linear Lagrange method reproduced the main nodal structure and overall shape of the oscillator wavefunctions, while also giving reasonable energy approximations. Its accuracy was limited by the piecewise-linear nature of the basis, especially for higher states with more oscillatory structure.

The cubic Hermite method achieved smaller

errors with fewer elements. This behavior follows from its higher-order C^1 approximation space, which uses both wavefunction values and derivative degrees of freedom. For the smooth harmonic oscillator eigenfunctions, these additional degrees of freedom allow the numerical solution to represent curvature more efficiently.

The convergence study supports this interpretation. Under mesh refinement, the linear P_1 method displayed approximately second-order convergence, while the cubic Hermite method displayed approximately fourth-order convergence. The observed order difference explains why the Hermite discretization reaches higher accuracy with fewer elements.

The domain truncation study also identified an important practical constraint. The finite interval $[-L, L]$ must be large enough to contain the decaying wavefunction tails. At a fixed element count, however, increasing L also increases the element size and reduces resolution in the region of significant amplitude. Accurate finite element solutions require both appropriate boundary placement and adequate mesh resolution.

Overall, the finite element method gives an effective variational approximation for this quantum bound-state problem. Both implemented discretizations recover the expected spectrum and wavefunction shapes. The cubic Hermite method is more accurate for this smooth problem because its higher-order, derivative-based approximation space is better matched to the regularity of the exact oscillator states. This conclusion applies to the implementation studied here and should be understood as a comparison

of the two discretizations used in this report.

REFERENCES

- [1] L. R. Ram-Mohan, *Finite Element and Boundary Element Applications in Quantum Mechanics*. Oxford University Press, 2002.